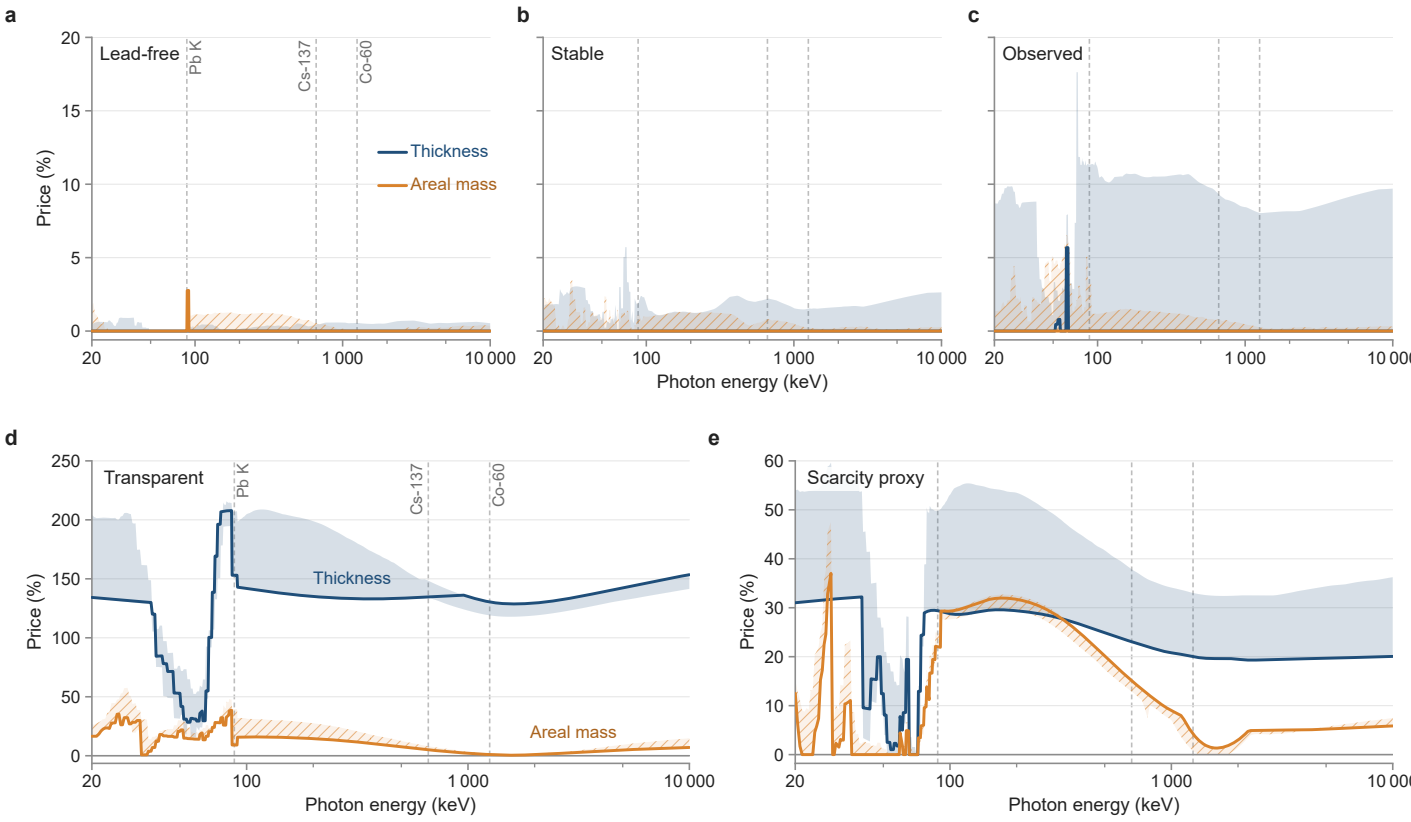

a
Lead-free
Pb K
Cs-137
Co-60
Thickness
Areal mass
Price (%)
20
15
10
5
0
20
100
1 000
10 000
b
Stable
c
Observed
Photon energy (keV)
d
Transparent
Pb K
Cs-137
Co-60
Thickness
Areal mass
Price (%)
250
200
150
100
50
0
Photon energy (keV)
e
Scarcity proxy
Price (%)
60
50
40
30
20
10
0
Photon energy (keV)